DDR2 – Why Consider It?

Feature/Option	DDR	DDR2	DDR2 Advantage
Package	TSOP (66 pins)	FBGA only	**Enables better electrical performance and speed**
Voltage	2.5V 2.5V I/O	1.8V 1.8V I/O	**Reduces memory system power demand**
Densities	128Mb–1Gb	256Mb–2Gb	**High-density components enable large memory subsystems**
Internal Banks	4	4 and 8	**1Gb and higher DDR2 devices will have 8 banks for better performance**
Prefetch (MIN WRITE Burst)	2	4	**Provides reduced core speed dependency for better yields**
Speed (Data Pin)	200 MHz, 266 MHz, 333 MHz, and 400 MHz	400 MHz, 533 MHz, and 667 MHz	**Migration to higher bus speed**
READ Latency ***Additive Latency (Posted CAS)***	2, 2.5, 3 CLK N/A	CL + AL CL = 3, 4, 5 AL options 0, 1, 2, 3, 4	**Eliminating one-half clock settings helps speed internal DRAM logic and improves yields** **Mainly used in server applications to improve command bus efficiency**
WRITE Latency	1 clock	READ latency - 1	**Improves command bus efficiency**
Termination	Motherboard parallel to VTT	DRAM on-die termination (ODT), optional on-motherboard termination	**ODT for both memory and controller improves signaling and reduces system cost**
Data Strobes	Single-ended	Differential or single-ended	**Improves system timing margin by reducing strobe crosstalk**
Modules	184-pin DIMM unbuffered registered 200-pin SODIMM 172-pin MicroDIMM	240-pin DIMM unbuffered registered 200-pin SODIMM 244-pin MicroDIMM	**Improved layout and power delivery design**
Chipset Support	All DTs, NBs, and servers	All DTs, NBs, and servers	**All major chipset providers**

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